

Abstract

Examine network traffic to determine what information is exposed in transit. Sample traffic is generated and sent. The packets are observed and examined for content. The results demonstrate that all information sent is clear to read by an attacker.

Introduction

Kali Linux will be used in a VirtualBox Virtual machine. Netcat will be used to send and receive unencrypted data. Wireshark will be used for packet analysis.

Netcat listener commands used:

```
nc -l 127.0.0.1 -p 31337
```

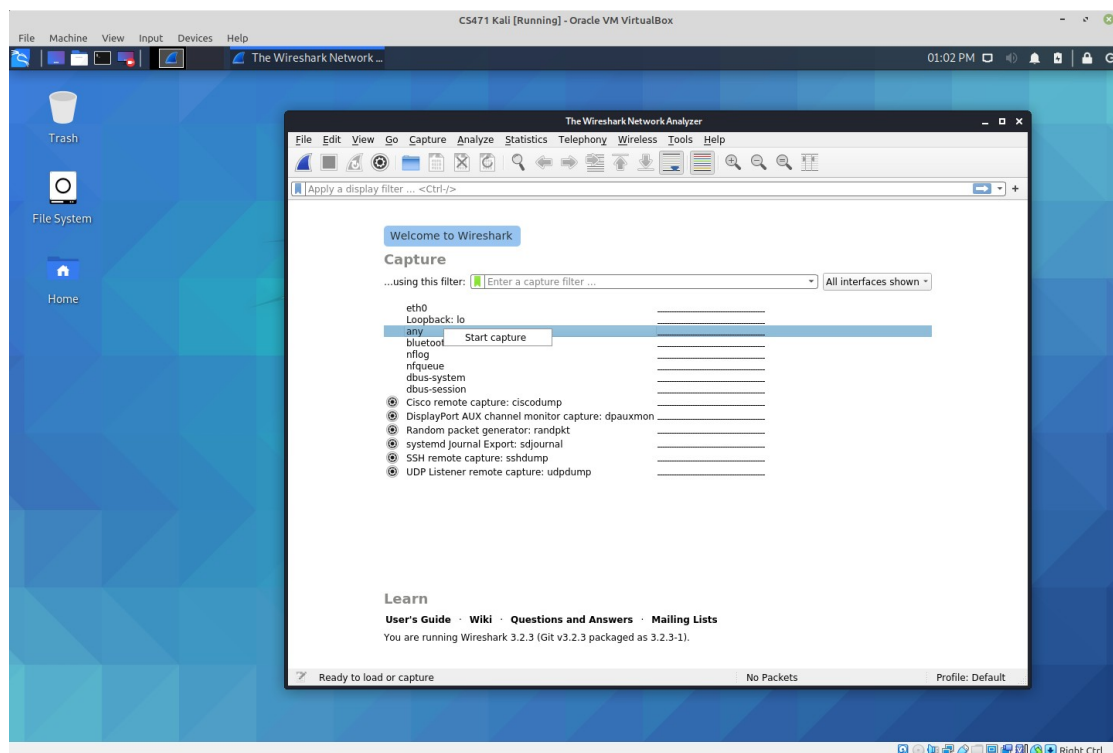
Netcat sender commands:

```
nc 127.0.0.1 -p 31337
```

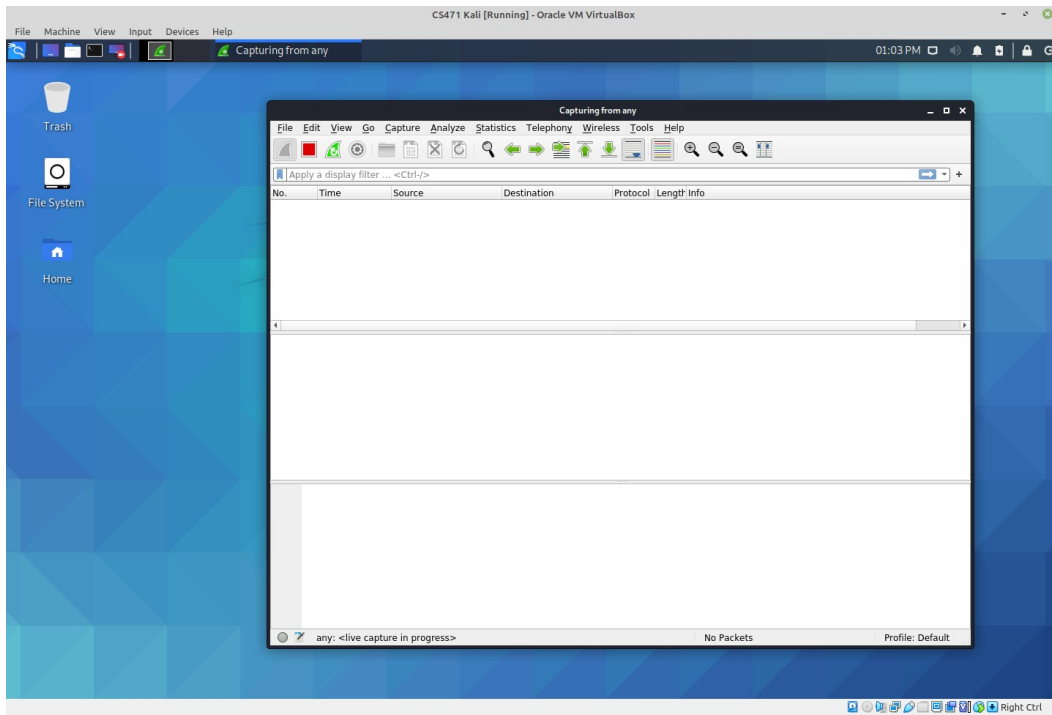
Wireshark run as root while capturing on the 'any' adapter.

Summary of Results

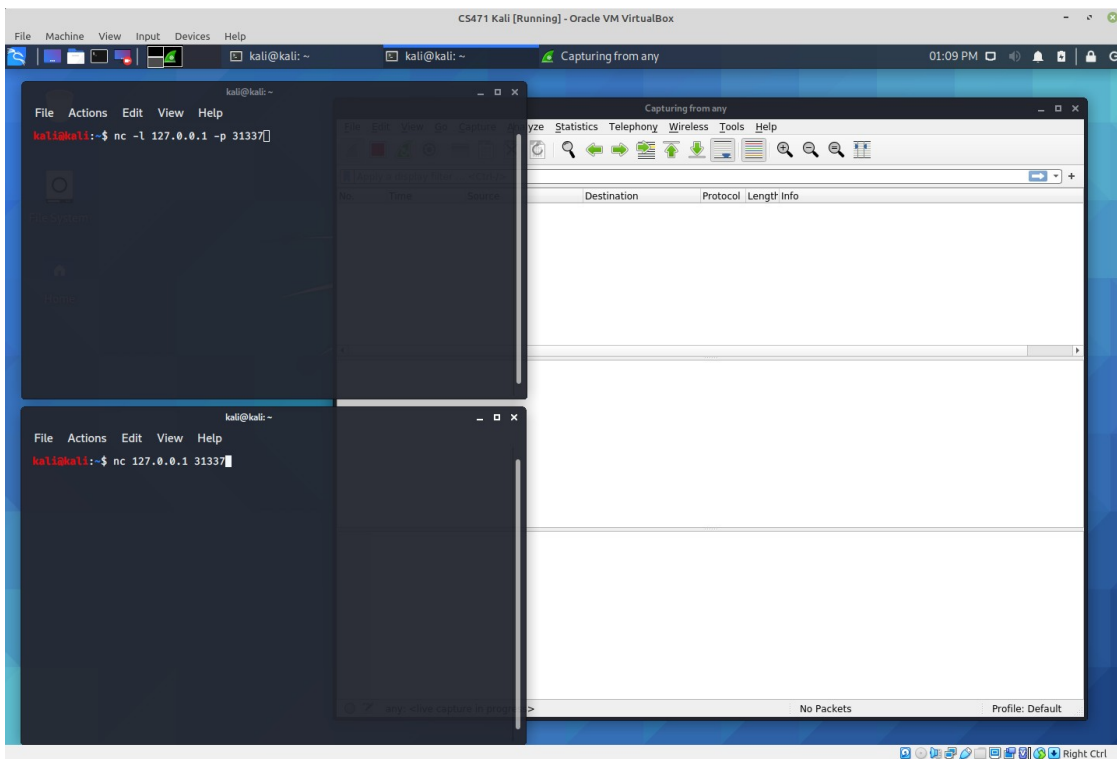
Starting Wireshark in Kali Linux. Selecting the 'any' interface for packet capture.



Now Wireshark is listening. No packets heard.



Next open two terminals. One as a netcat listener first, then another as a Netcat sender.



The image shows a Kali Linux virtual machine environment. On the left, a terminal window displays a netcat listener on port 31337, which has successfully connected to 127.0.0.1. On the right, the Wireshark network protocol analyzer is running, capturing traffic on the 'any' interface. The packet list shows three captured packets, all of which are TCP connections from 127.0.0.1 to 127.0.0.1 on port 31337. The packet details pane for the selected packet (No. 3) shows the following structure:

- Frame 1: 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface any, id 0
- Linux cooked capture
- Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- Transmission Control Protocol, Src Port: 54808, Dst Port: 31337, Seq: 0, Len: 0

The packet bytes pane shows the raw data of the captured packet:

```
0000  00 00 03 04 00 06 00 00 00 00 00 00 00 00 00 00  .....
0010  45 00 00 3c 42 0d 40 00 40 06 fa ac 7f 00 00 01  E...B...:.....
0020  7f 00 00 01 06 18 7a 69 c1 a7 64 43 00 00 00 00  ....Z1..dC.....
0030  a0 02 ff 07 fe 30 00 00 02 04 ff 07 04 02 08 0a  .....
0040  b7 e4 12 98 00 00 00 01 03 03 07  .....

```

The image shows a Kali Linux virtual machine running Oracle VM VirtualBox. The terminal window displays a netcat listener on 127.0.0.1:31337, which receives a connection from 127.0.0.1. The Wireshark network analyzer is open, capturing traffic on the 'any' interface. It shows a TCP connection established between 127.0.0.1 and 127.0.0.1 on port 31337. The packet list shows a SYN packet, followed by an ACK packet, and then a data packet. The packet details pane shows the TCP segment structure, including the sequence number, acknowledgment number, and window size. The packet bytes pane shows the raw data being transmitted.

Terminal Output:

```
kali@kali:~$ nc -l 127.0.0.1 -p 31337
This is a test

kali@kali:~$ nc 127.0.0.1 31337
This is a test
```

Wireshark Capture:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	60	45088 → 31337 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 S
2	0.000013191	127.0.0.1	127.0.0.1	TCP	76	31337 → 54808 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 W
3	0.000023939	127.0.0.1	127.0.0.1	TCP	68	54808 → 31337 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=
4	83.501267602	127.0.0.1	127.0.0.1	TCP	83	54808 → 31337 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=15
5	83.501302912	127.0.0.1	127.0.0.1	TCP	68	31337 → 54808 [ACK] Seq=1 Ack=16 Win=65536 Len=0 TSval=

Packet Details:

Frame 1: 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface any, id 0

Linux cooked capture

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 54808, Dst Port: 31337, Seq: 0, Len: 0

Packet Bytes:

```
00 00 00 00 00 00 00 00 .....
06 fa ac 7f 00 00 01 E..<B>00.....
47 64 43 00 00 00 01 .....21...dC.....
04 ff 07 04 02 08 0a .....
03 03 07 .....
```

Notice the text sent over the network was found in the Wireshark capture. The text was easily intercepted and revealed.

CS471 Kali [Running] - Oracle VM VirtualBox

FileMachineViewInputDevicesHelp

kali@kali: ~kali@kali: ~Capturing from any01:14 PM

Capturing from any

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000...	127.0.0.1	127.0.0.1	TCP	76	54808 → 31337 [SYN] Seq=0 Win=6...
2	0.000...	127.0.0.1	127.0.0.1	TCP	76	31337 → 54808 [SYN, ACK] Seq=0 ...
3	0.000...	127.0.0.1	127.0.0.1	TCP	68	54808 → 31337 [ACK] Seq=1 Ack=1...
4	83.50...	127.0.0.1	127.0.0.1	TCP	83	54808 → 31337 [PSH, ACK] Seq=1 ...
5	83.50...	127.0.0.1	127.0.0.1	TCP	68	31337 → 54808 [ACK] Seq=1 Ack=1...

Frame 4: 83 bytes on wire (664 bits), 83 bytes captured (664 bits) on interface any, id 0

Linux cooked capture

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 54808, Dst Port: 31337, Seq: 1, Ack: 1, Len: 15

Data (15 bytes)

Data: 54686973206973206120746573740a

[Length: 15]

0000	00 00 03 04 00 06 00 00	00 00 00 00 00 00 08 00	
0010	45 00 00 43 42 0f 40 00	40 06 fa a3 7f 00 00 01	E..CB.@. @.....
0020	7f 00 00 01 d6 18 7a 69	c1 a7 64 44 15 3c 5b d2	zi ..dD.<[.
0030	80 18 02 00 fe 37 00 00	01 01 08 0a b7 e5 58 c5	7.....X.
0040	b7 e4 12 98 54 68 69 73	20 69 73 20 61 20 74 65	...This is a te
0050	73 74 0a		st.

Data (data.data), 15 bytes

Packets: 5 · Displayed: 5 (100.0%)

Profile: Default

Right Ctrl

Conclusion

Packets sent over the network are clear to read by any device that intercepts them. Additional measures are required to secure private information.

Note to students:

This sample report is an example of the *format* to use for assignment submission; it lacks sufficient details for a full credit submission. Be sure to include more substance than this example. For each submitted screenshot, be sure to provide a short paragraph explaining the activity shown.

The report should speak to a technical audience and provide enough details so your work is reproducible by following the documented steps. The body of your report should clearly demonstrate your progress when completing the assignment. This includes any difficulties you may have experienced along the way.

Be sure to document your results. Use the conclusion section to explain your results.

Abstract: The abstract is a very short executive summary of the entire document. This includes result. 3-4 sentences only.

Introduction: This includes the tools used and any background information needed to understand your report.

Summary of Results: This includes your work in a *reproducible* way. Explain all of your work here. Include screenshots with descriptions. There should be more words here than pictures....

Conclusion: This should be a cogent explanation of the results. This should explain what the results imply and what the importance of this is. If prompted for a specific conclusion topic, include this, but do not exclude your explanation of results.

Last, try to avoid editorializing and using marketing jargon. Be clear and support your claims with results.